A 1-MORPHISM OF STACKS IS FULLY FAITHFUL IFF IT INDUCES ISOMORPHISMS OF MOR SHEAVES

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Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Strongly cartesian lifting. Categories, Section fibred-categories (cross-chapter, named). Provides the lifting f^*y -> y used to reduce morphisms over an arbitrary arrow to morphisms in one fibre.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $\mathcal{S}_1$, $\mathcal{S}_2$ be stacks over $\mathcal{C}$. Let $F : \mathcal{S}_1 \to \mathcal{S}_2$ be a 1-morphism. Then the following are equivalent*

(1) *F is fully faithful,*
(2) *for every $U \in \mathrm{Ob}(\mathcal{C})$ and for every $x, y \in \mathrm{Ob}(\mathcal{S}_{1,U})$ the map*
$$F : Mor_{\mathcal{S}_1}(x, y) \longrightarrow Mor_{\mathcal{S}_2}(F(x), F(y))$$
is an isomorphism of sheaves on $\mathcal{C}/U$.

Proof. Assume (1). For U, x, y as in (2) the displayed map F evaluates to the map $F : \mathrm{Mor}_{\mathcal{S}_{1,V}}(x|_V, y|_V) \to \mathrm{Mor}_{\mathcal{S}_{2,V}}(F(x|_V), F(y|_V))$ on an object V of $\mathcal{C}$ lying over U. Now, since F is fully faithful, the corresponding map $\mathrm{Mor}_{\mathcal{S}_1}(x|_V, y|_V) \to \mathrm{Mor}_{\mathcal{S}_2}(F(x|_V), F(y|_V))$ is a bijection. Morphisms in the fibre category $\mathcal{S}_{1,V}$ are exactly those morphisms between $x|_V$ and $y|_V$ in $\mathcal{S}_1$ lying over id_V. Similarly, morphisms in the fibre category $\mathcal{S}_{2,V}$ are exactly those morphisms between $F(x|_V)$ and $F(y|_V)$ in $\mathcal{S}_2$ lying over id_V. Thus we find that F induces a bijection between these also. Hence (2) holds.

Assume (2). Suppose given objects U, V of $\mathcal{C}$ and $x \in \mathrm{Ob}(\mathcal{S}_{1,U})$ and $y \in \mathrm{Ob}(\mathcal{S}_{1,V})$. To show that F is fully faithful, it suffices to prove it induces a bijection on morphisms lying over a fixed $f : U \to V$. Choose a strongly Cartesian $f^*y \to y$ in $\mathcal{S}_1$ lying above f. This results in a bijection between the set of morphisms $x \to y$ in $\mathcal{S}_1$ lying over f and $\mathrm{Mor}_{\mathcal{S}_{1,U}}(x, f^*y)$. Since F preserves strongly Cartesian morphisms as a 1-morphism in the 2-category of stacks over $\mathcal{C}$, we also get a bijection between the set of morphisms $F(x) \to F(y)$ in $\mathcal{S}_2$ lying over f and $\mathrm{Mor}_{\mathcal{S}_{2,U}}(F(x), F(f^*y))$. Since F induces a bijection $\mathrm{Mor}_{\mathcal{S}_{1,U}}(x, f^*y) \to \mathrm{Mor}_{\mathcal{S}_{2,U}}(F(x), F(f^*y))$ we conclude (1) holds. □

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